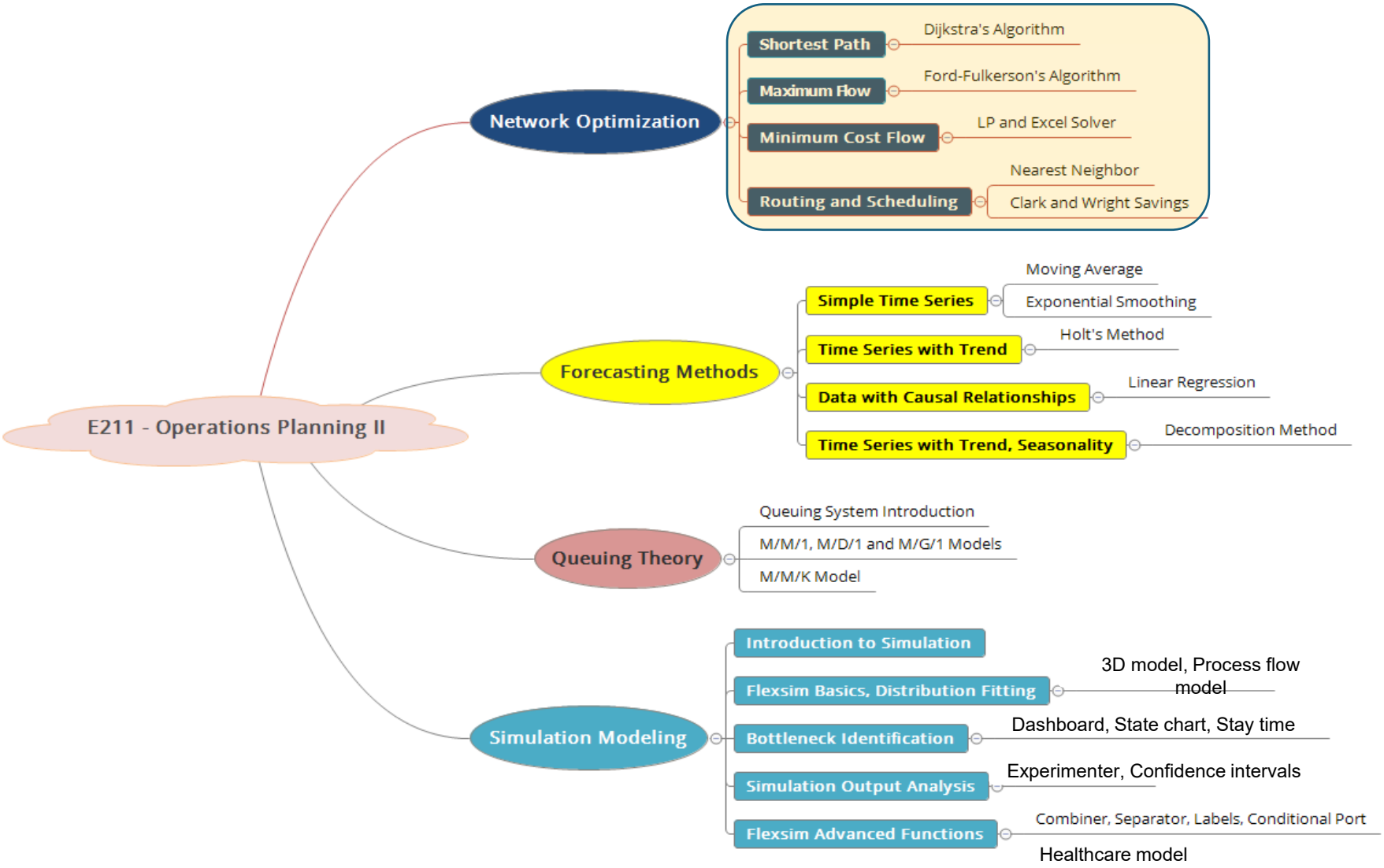
Lessons 16, 18 & 20 Solution

E211 – Operations Planning II

Module Coverage: E211 Topic Tree



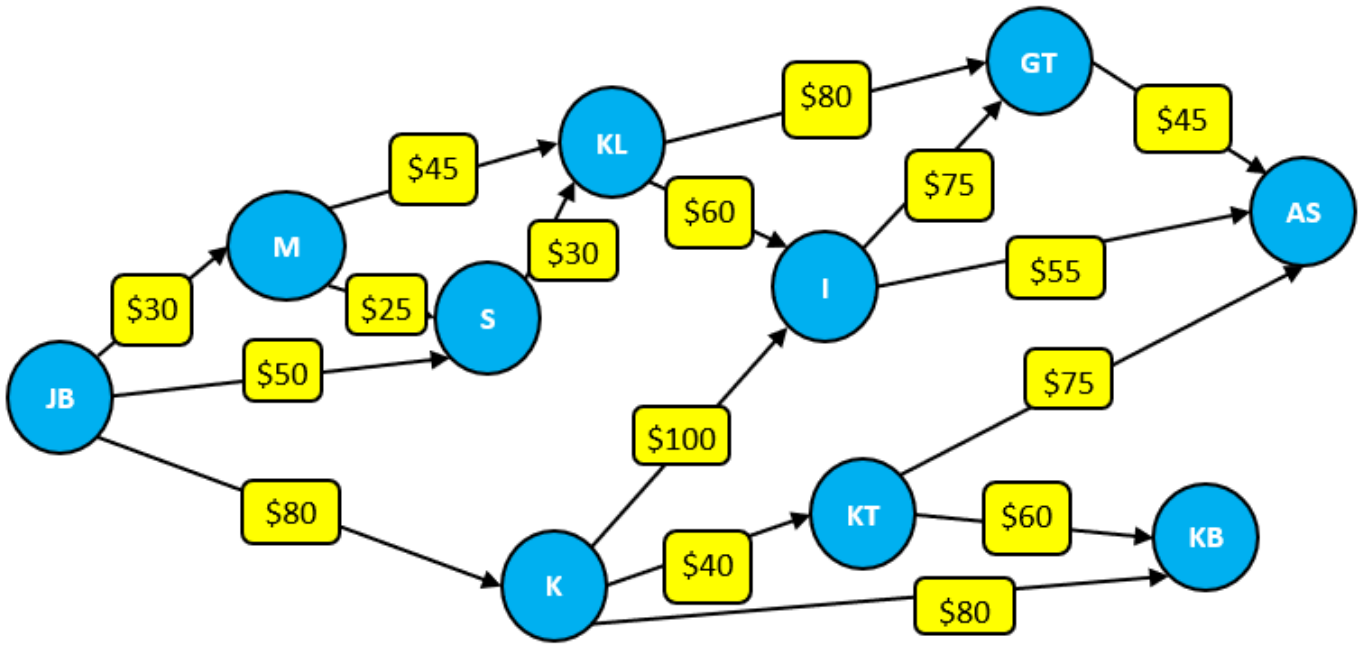
Lesson 16
Solution

Shortest path problem & Maximum flow problem

Learning Outcomes

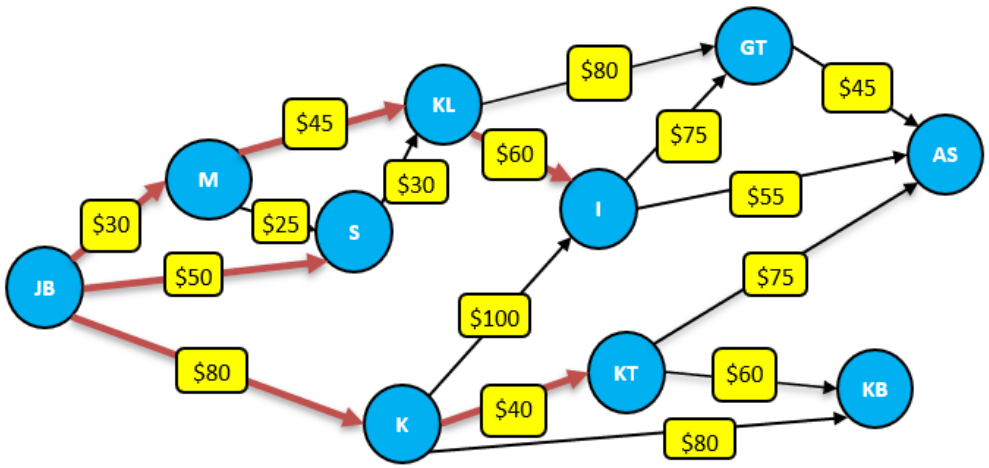
- Identify a Shortest Path problem.
- Apply Dijkstra's Algorithm to solve a Shortest Path problem.
- Identify a Maximum Flow problem.
- Apply Ford-Fulkerson's Algorithm to solve a Maximum Flow problem.

The Shortest Path Problem



- The delivery routes can be represented as a network model.
- The ten cities are represented by nodes.
- The delivery routes are represented by arcs.
- Minimum cost delivery route from Johor Bahru to each destination can be determined by solving as a shortest path problem either using the Dijkstra's Algorithm or LP model.

Dijkstra's Algorithm



Iteration	Solution Set	Arcs	Total Cost
Iteration 1	JB	**JB - M** JB - S JB - K	**30** 50 80

Iteration	Solution Set	Arcs	Total Cost
Iteration 2	JB M (30)	**JB - S** JB - K M - S M - KL	**50** 80 30+25 = 55 30+45 = 75

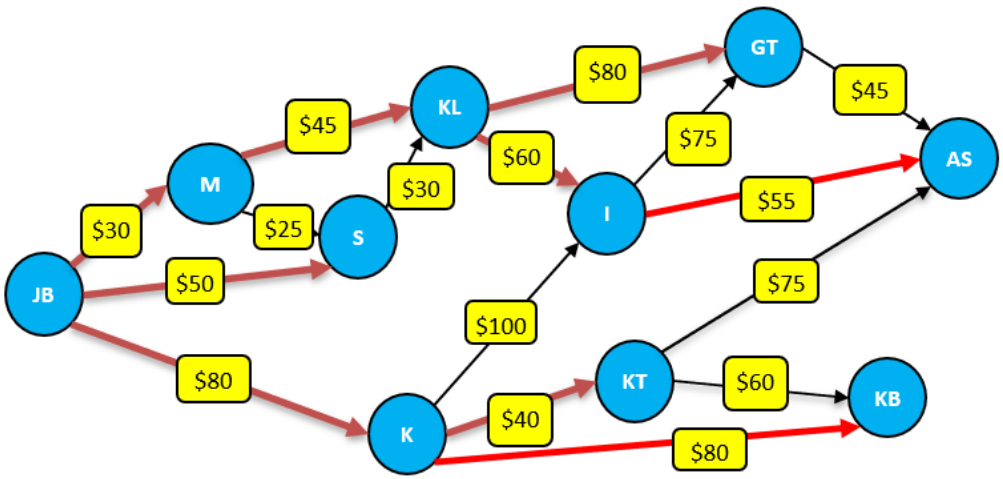
Iteration	Solution Set	Arcs	Total Cost
Iteration 3	JB M (30) S (50)	JB -> K **M -> KL** S -> KL	80 **75** 50+30 = 80

Iteration	Solution Set	Arcs	Total Cost
Iteration 4	JB M (30) S (50) KL (75)	**JB - K** KL - I KL - GT	**80** 75+60 = 135 75+80 = 155

Iteration	Solution Set	Arcs	Total Cost
Iteration 5	JB M (30) S (50) KL (75) K (80)	KL - I KL - GT **K - KT** K - I K - KB	75+60 = 135 75+80 = 155 **80+40 = 120** 80+100 = 180 80+80 = 160

Iteration	Solution Set	Arcs	Total Cost
Iteration 6	JB M (30) S (50) KL (75) K (80) KT (120)	**KL - I** KL - GT K - I K - KB KT - KB KT - AS	**75+60 = 135** 75+80 = 155 80+100 = 180 80+80 = 160 120+60 = 180 120+75 = 195

Dijkstra's Algorithm



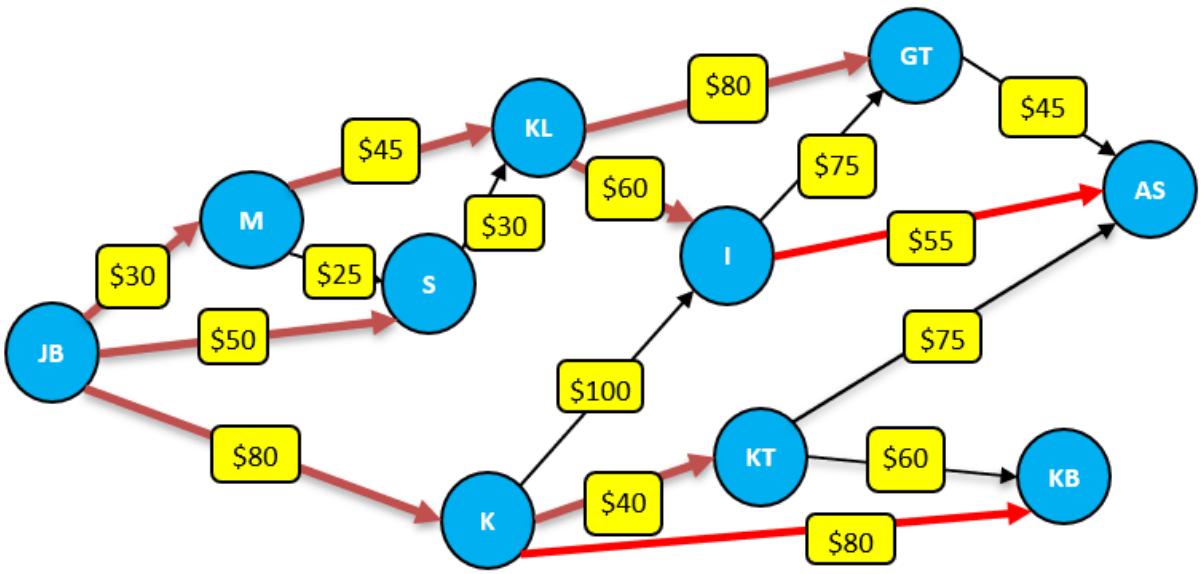
Iteration	Solution Set	Arcs	Total Cost
Iteration 7	JB M (30) S (50) KL (75) K (80) KT (120) I (135)	**KL - GT** K - KB KT - KB KT - AS I - AS I - GT	**75+80 = 155** 80+80 = 160 120+60 = 180 120+75 = 195 135+65 = 200 135+75 = 210

Iteration	Solution Set	Arcs	Total Cost
Iteration 8	JB M (30) S (50) KL (75) K (80) KT (120) I (135) GT (155)	**K - KB** KT - KB KT - AS I - AS GT - AS	**80+80 = 160** 120+60 = 180 120+75 = 195 135+55 = 190 155+45 = 200

Iteration	Solution Set	Arcs	Total Cost
Iteration 9	JB M (30) S (50) KL (75) K (80) KT (120) I (135) GT (155) KB (160)	KT - AS **I - AS** GT - AS	120+75 = 195 **135+55 = 190** 155+45 = 200

Iteration	Solution Set	Arcs	Total Cost
Iteration 10	JB M (30) S (50) KL (75) K (80) KT (120) I (135) GT (155) KB (160) AS (190)		

The Shortest Path Solution

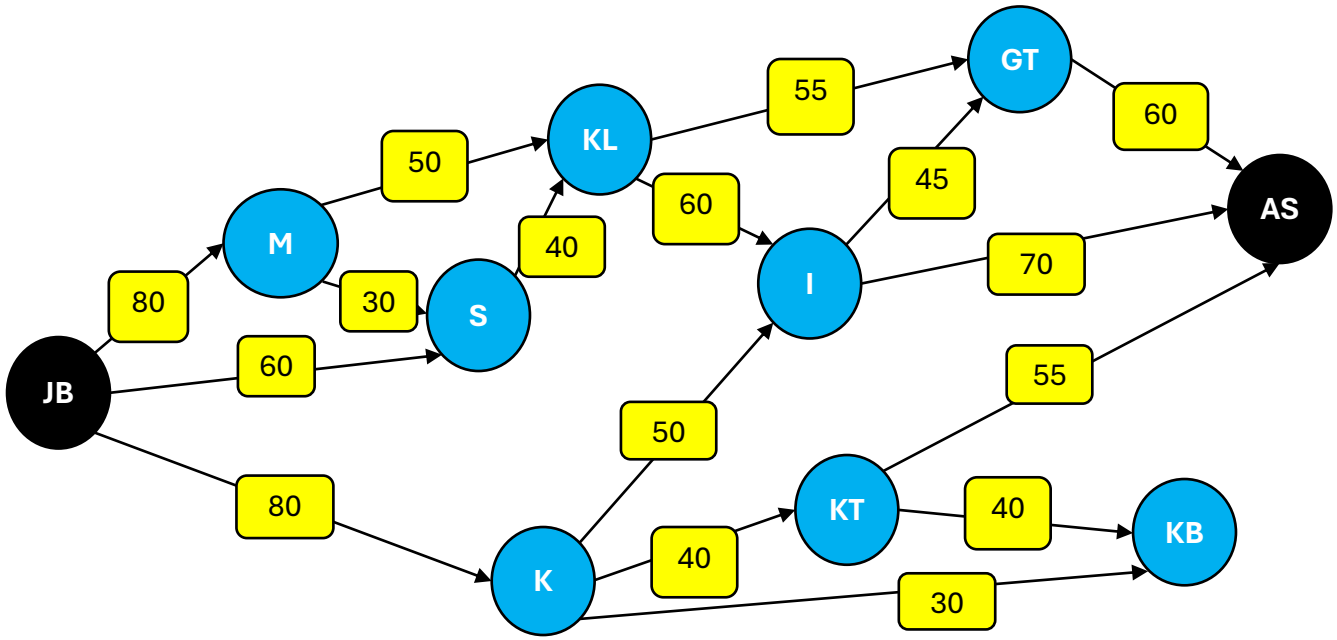


From JB to	Shortest Path	Total Cost
M	JB -> M	30
S	JB -> S	50
KL	JB -> M -> KL	75
K	JB -> K	80
KT	JB -> K -> KT	120
I	JB -> M -> KL -> I	135
GT	JB -> M -> KL -> GT	155
KB	JB -> K -> KB	160
AS	JB -> M -> KL -> I -> AS	190

<u>Assumptions</u>

- Unit cost of shipment does not change with shipment quantity.
- Routes in the network are fixed.
- There is no limitation of route capacity.

The Maximum Flow Problem



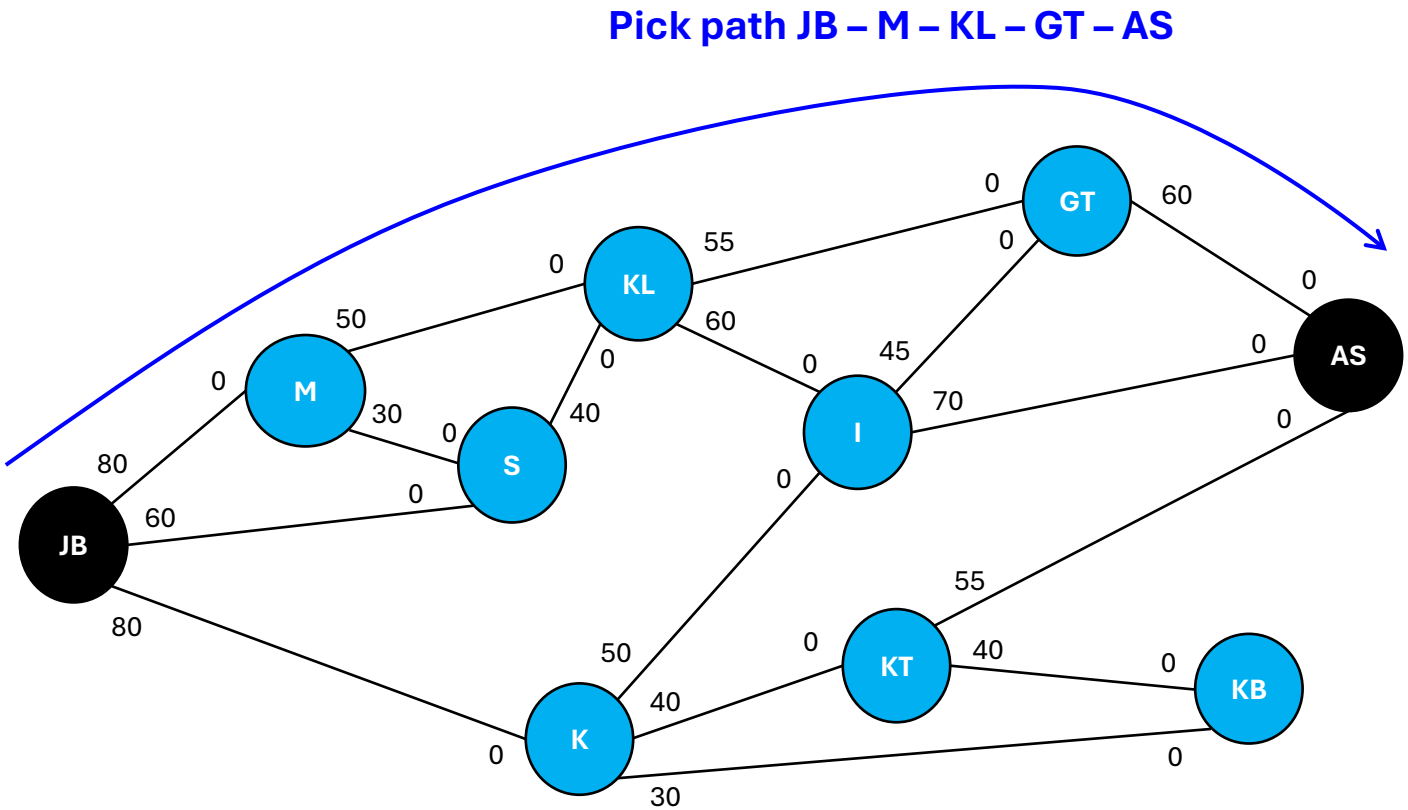
- To determine the maximum flow of products from Johor Bahru (Origin) to Alor Setar (Destination), and the amount to deliver along each route, we apply Ford-Fulkerson's Algorithm.

- Assumptions:
  - Cost of delivery is not a factor in determining the routes
  - Products are only delivered from Johor to Alor Setar
  - Delivery routes and capacity are fixed

Ford Fulkerson's Algorithm

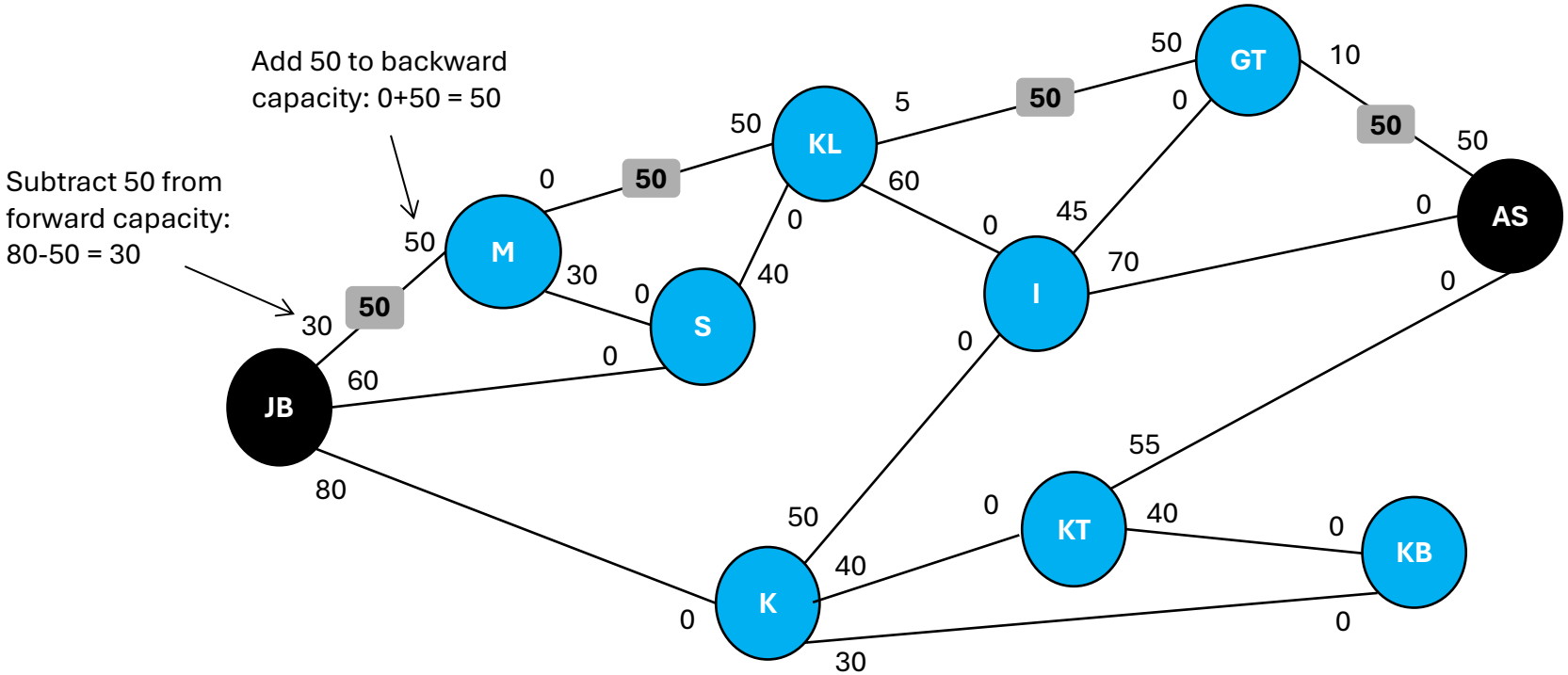
Solution Step 1

Convert routes to non-directional arcs and write down forward and reverse arc capacities.



Ford Fulkerson's Algorithm

Solution Step 2: Work out Flow Quantity and Balance Capacities



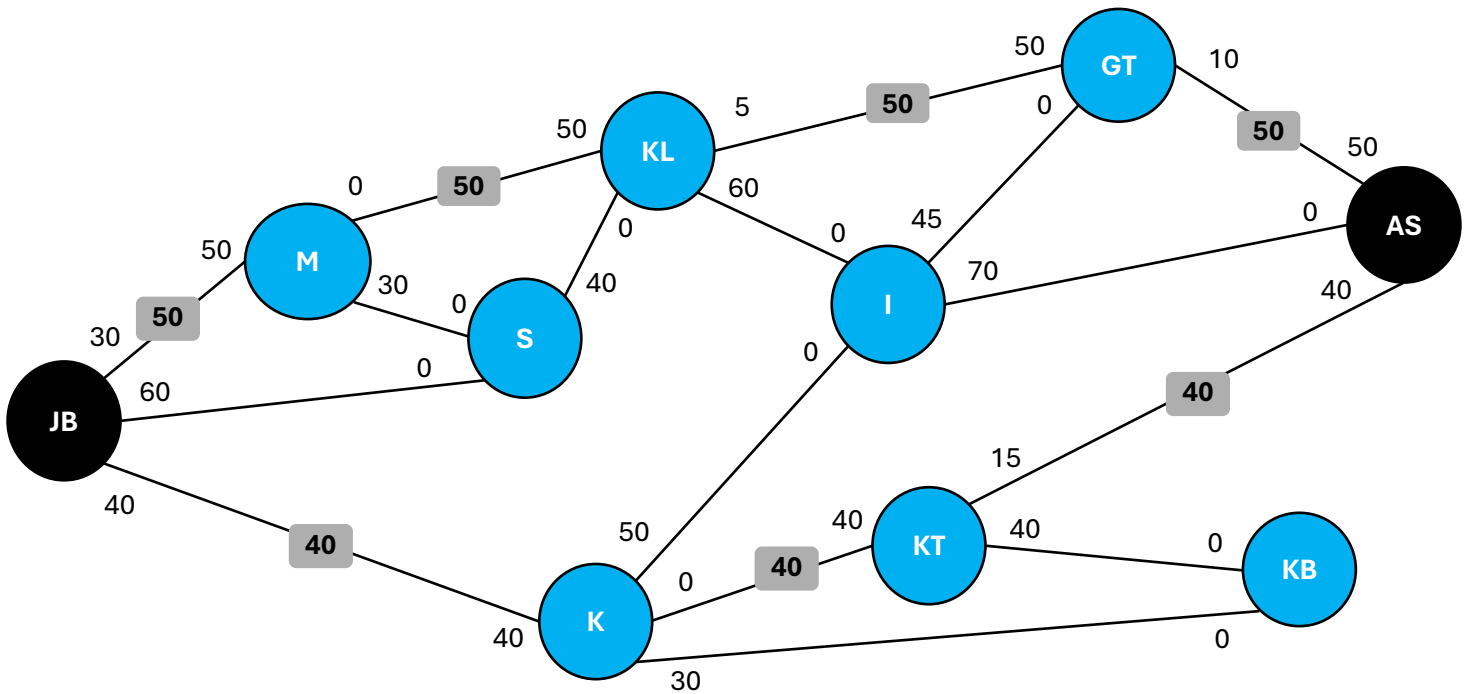
Route	Flow Quantity
JB - M - KL - GT - AS	50

Flow quantity = **smallest** capacity of all arcs along the route

Next, pick path JB – K – KT – AS

Ford Fulkerson's Algorithm

Solution Step 3: Repeat workings for Path JB – K – KT – AS

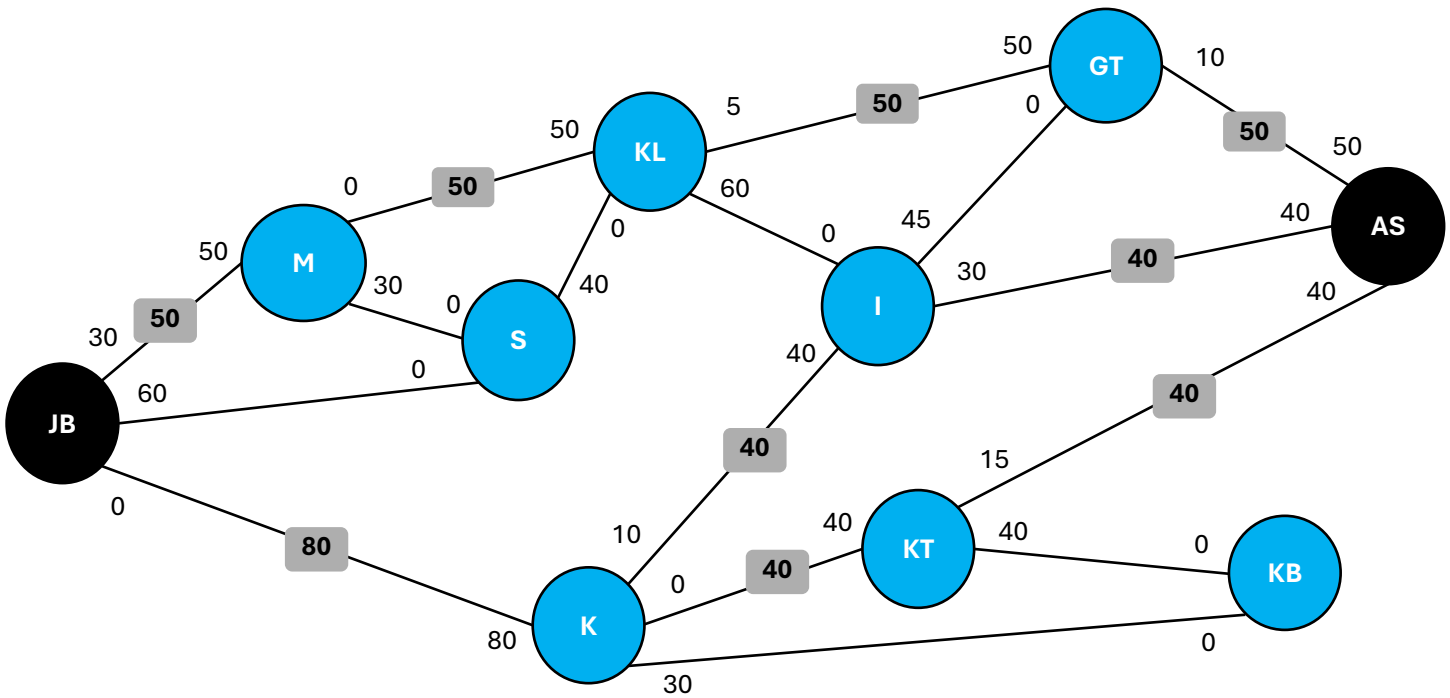


Route	Flow Quantity
JB - M - KL - GT - AS	50
JB - K - KT - AS	40

Next, pick path JB – K – I – AS

Ford Fulkerson's Algorithm

Solution Step 4: Repeat workings for Path JB – K – I – AS

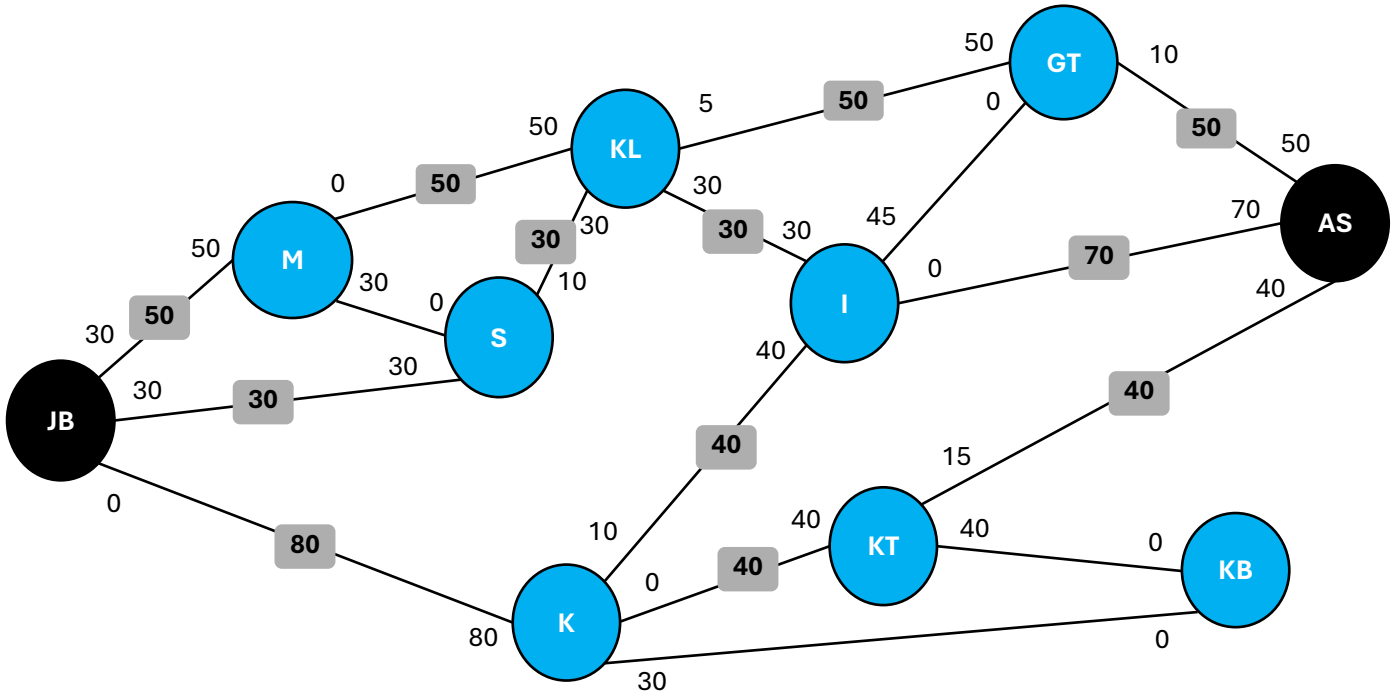


Route	Flow Quantity
JB - M - KL - GT - AS	50
JB - K - KT - AS	40
JB - K - I - AS	40

Next, pick path JB – S – KL – I – AS

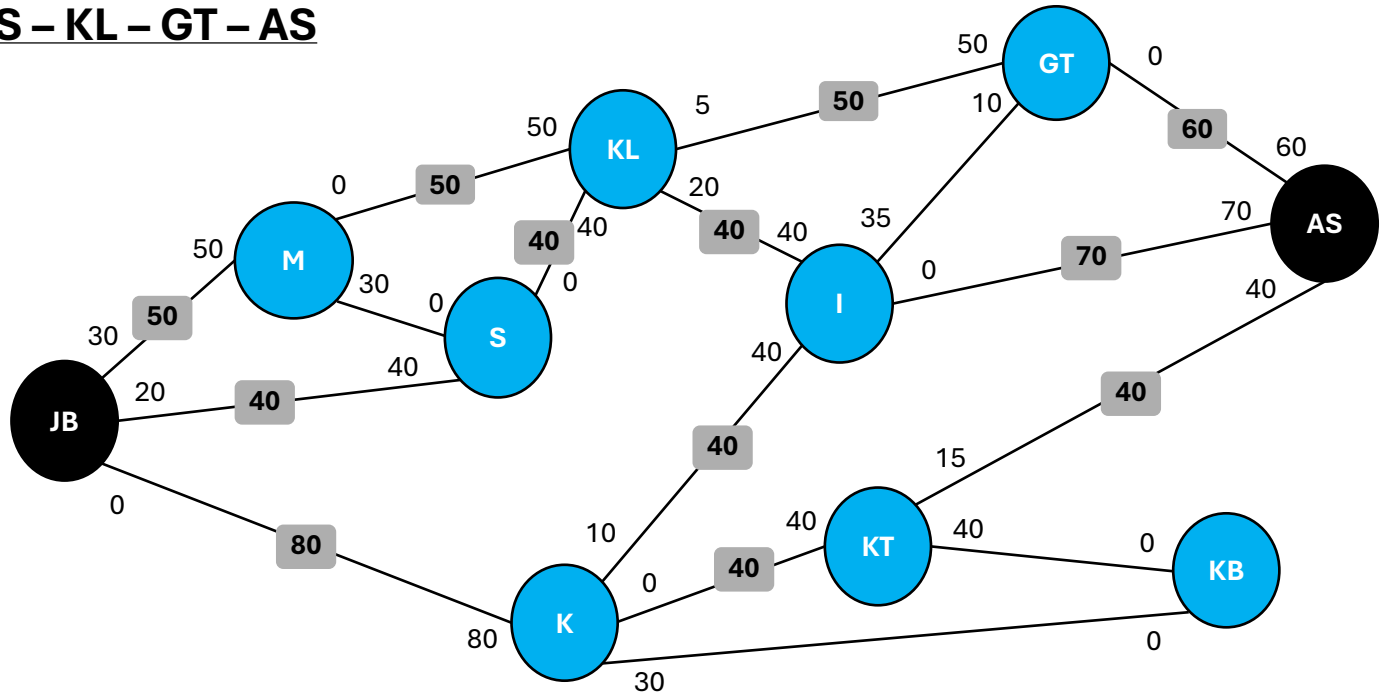
Ford Fulkerson's Algorithm

Solution Step 5: Repeat workings for Path JB – S – KL – I – AS


JB
M
S
KL
GT
I
AS
K
KT
KB
50
30
30
80
50
50
30
30
30
50
40
40
40
70
40
0
0
0

Ford Fulkerson's Algorithm

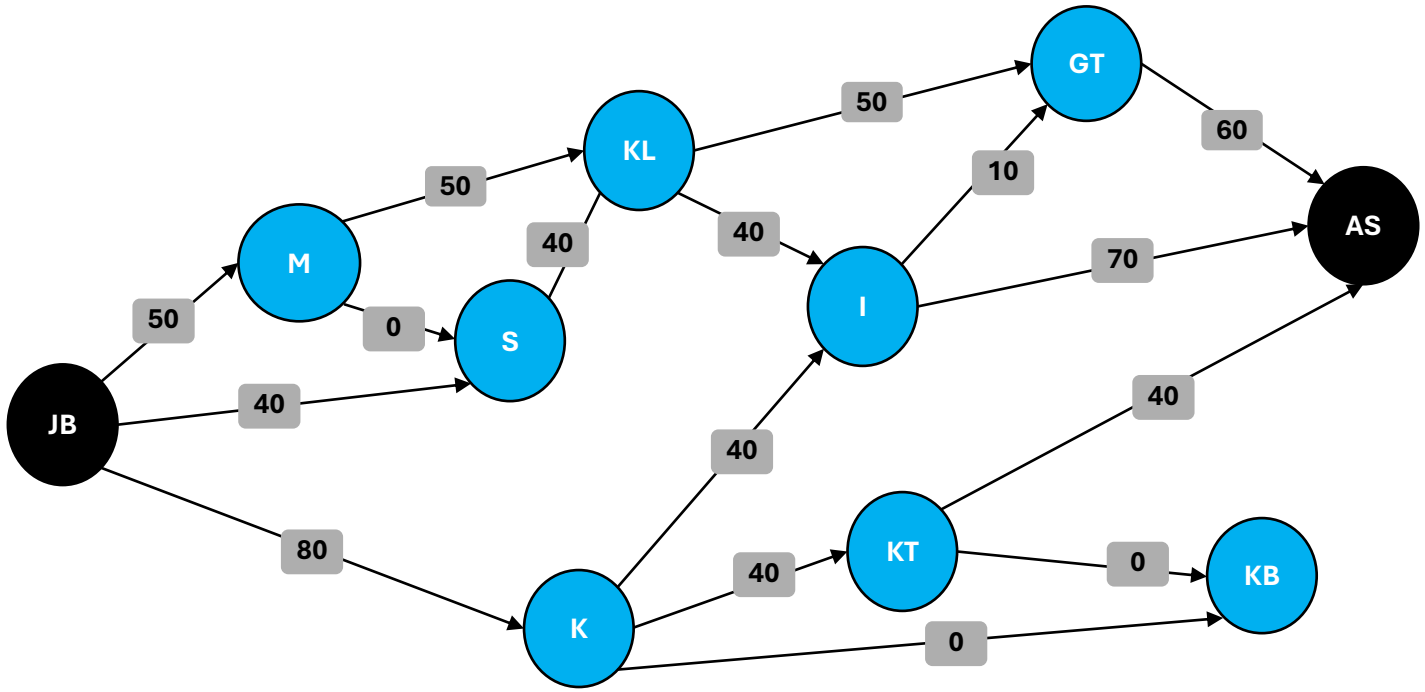
Solution Step 6: Repeat workings for Path JB – S – KL – GT – AS



Route	Flow Quantity
JB - M - KL - GT - AS	50
JB - K - KT - AS	40
JB - K - I - AS	40
JB - S - KL - I - AS	30
JB - S - KL - GT - AS	10

At this stage, there is no other path with capacity that is available.

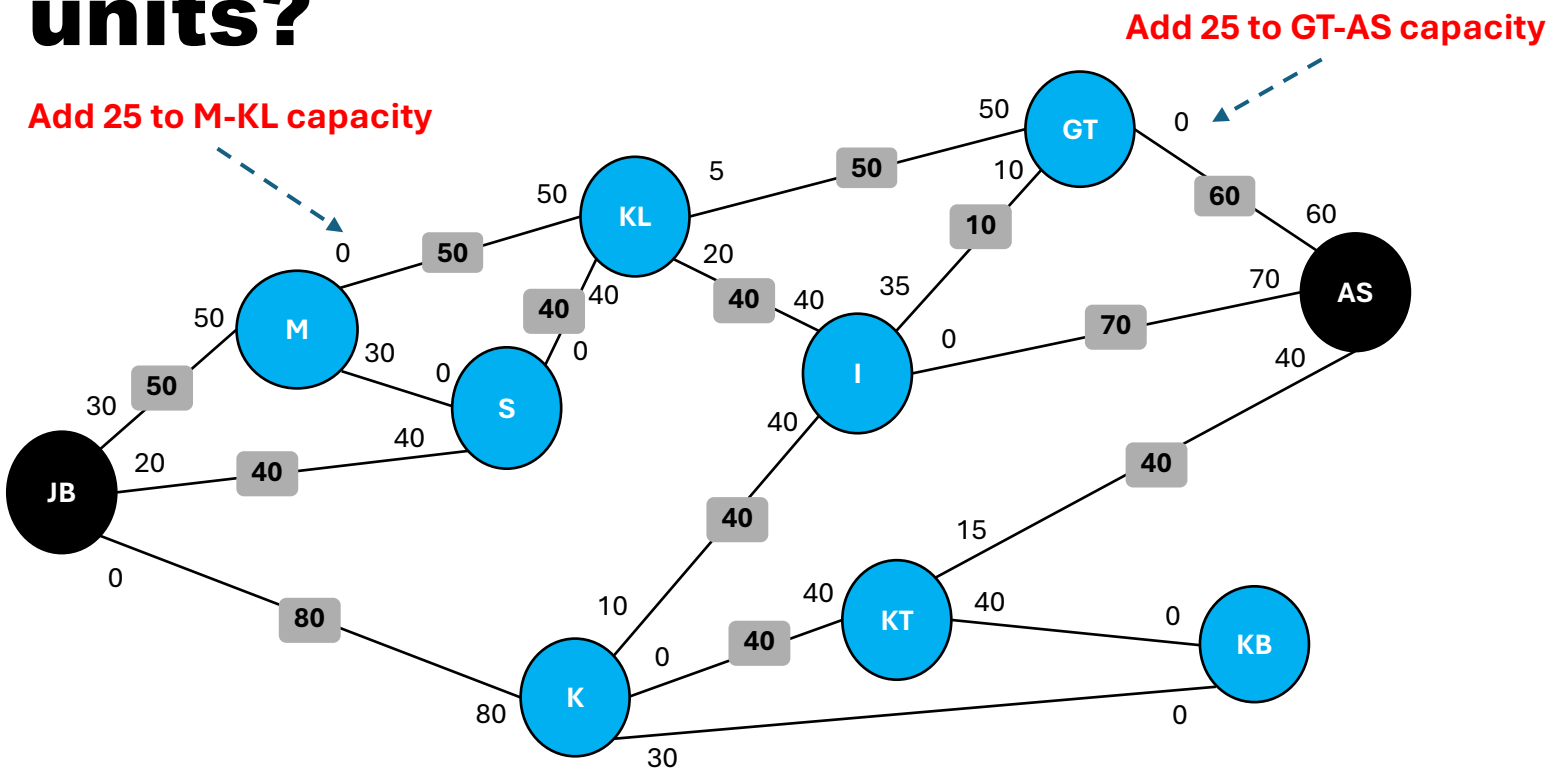
The Maximum Flow Solution



Route	Flow Quantity
JB - M - KL - GT - AS	50
JB - K - KT - AS	40
JB - K - I - AS	40
JB - S - KL - I - AS	30
JB - S - KL - GT - AS	10
Max Flow	**170**

There are multiple optimal solutions for this problem since the solution depends on selection of routes in the algorithm.

How to increase max flow by 25 units?



Increase max flow by 25 by increasing the capacity of:

- Arc M-KL by 25
- Arc GT-AS by 25

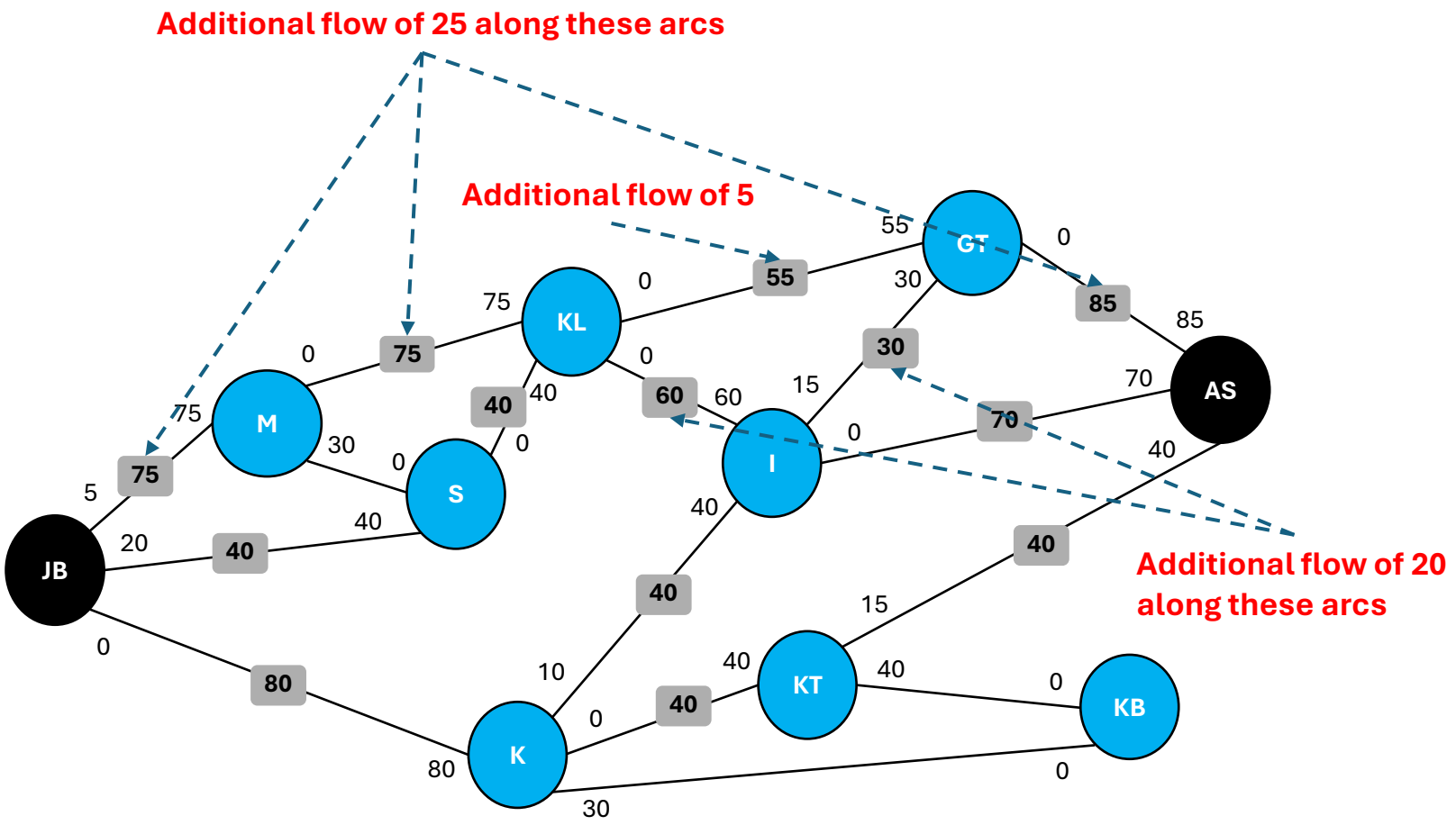
Additional network flow qty of 25 via:

JB - M - KL - GT - AS [5]

&

JB - M - KL - I - GT - AS [20]

Increase max flow by 25



Summary

- The cheapest delivery routes from Johor Bahru to each of the other 9 cities can be determined by solving it as a Shortest Path Problem.

- The maximum flow of products from Johor Bahru to Alor Setar and the amount of delivery along each route can be determined by solving it as a Maximum Flow Problem.

- Dijkstra's Algorithm and Ford-Fulkerson's Algorithm are used to solve Shortest Path and Maximum Flow Problem respectively. Both methods give optimal solutions.

Lesson 18 Solution

Nearest neighbour method and Clarke & Wright savings method

Learning Outcomes

- Identify what is a Vehicle Routing and Scheduling problem: Objectives, constraints, characteristics and classification of problems.

- Apply heuristic algorithm: Nearest Neighbour Method to obtain a feasible route.

- Apply heuristic algorithm: Clarke and Wright Savings method to obtain a feasible route.

The Nearest Neighbor Method Solution

From		To (Distance in km)								
		A	B	C	D	E	F	G	H	I
Serangoon North (Depot)	A	0	8.1	6	6	5.2	4.3	3.8	4	4
Jalan Angklong	B	8.1	0	13.5	14.4	14.3	9.1	6.7	9.5	9.3
Sumang link	C	6	13.5	0	3.7	3.6	9.1	9.7	6.2	5.6
Rivervale Crescent	D	6	14.4	3.7	0	1.7	6.8	7.1	4.4	4.2
Rivervale Street	E	5.2	14.3	3.6	1.7	0	6.5	7.1	4.2	4.5
Jalan Pelikat	F	4.3	9.1	9.1	6.8	6.5	0	2.3	7.7	8.2
Lorong Lew Lian	G	3.8	6.7	9.7	7.1	7.1	2.3	0	7.2	6.6
Fernvale Link	H	4	9.5	6.2	4.4	4.2	7.7	7.2	0	1.3
Sengkang West Way	I	4	9.3	5.6	4.2	4.5	8.2	6.6	1.3	0

From	To	Distance
A	G	3.8
G	F	2.3
F	E	6.5
E	D	1.7
D	C	3.7
C	I	5.6
I	H	1.3
H	B	9.5
B	A	8.1
	Total Dist.	42.5 km

Route:

A -> G -> F -> E -> D -> C -> I -> H -> B -> A

How will the solution change if the starting node is another node?

Clarke and Wright Savings Method

From		To (Distance in km)								
		A	B	C	D	E	F	G	H	I
Serangoon North (Depot)	A	0	8.1	6	6	5.2	4.3	3.8	4	4
Jalan Angklong	B	8.1	0	13.5	14.4	14.3	9.1	6.7	9.5	9.3
Sumang link	C	6	13.5	0	3.7	3.6	9.1	9.7	6.2	5.6
Rivervale Crescent	D	6	14.4	3.7	0	1.7	6.8	7.1	4.4	4.2
Rivervale Street	E	5.2	14.3	3.6	1.7	0	6.5	7.1	4.2	4.5
Jalan Pelikat	F	4.3	9.1	9.1	6.8	6.5	0	2.3	7.7	8.2
Lorong Lew Lian	G	3.8	6.7	9.7	7.1	7.1	2.3	0	7.2	6.6
Fernvale Link	H	4	9.5	6.2	4.4	4.2	7.7	7.2	0	1.3
Sengkang West Way	I	4	9.3	5.6	4.2	4.5	8.2	6.6	1.3	0

Clarke and Wright Savings Table

From		Savings								
		A	B	C	D	E	F	G	H	I
Serangoon North (Depot)	A	-	-	-	-	-	-	-	-	-
Jalan Angklong	B	-	-	0.6	-0.3	-1	3.3	5.2	2.6	2.8
Sumang link	C	-	0.6	-	8.3	7.6	1.2	0.1	3.8	4.4
Rivervale Crescent	D	-	-0.3	8.3	-	9.5	3.5	2.7	5.6	5.8
Rivervale Street	E	-	-1	7.6	9.5	-	3	1.9	5	4.7
Jalan Pelikat	F	-	3.3	1.2	3.5	3	-	5.8	0.6	0.1
Lorong Lew Lian	G	-	5.2	0.1	2.7	1.9	5.8	-	0.6	1.2
Fernvale Link	H	-	2.6	3.8	5.6	5	0.6	0.6	-	6.7
Sengkang West Way	I	-	2.8	4.4	5.8	4.7	0.1	1.2	6.7	-

Calculate the Savings for linking each pair of nodes (other than Depot (A) node)

Savings Calculation (Example): $S_{BC} = D_{A-B} + D_{A-C} - D_{B-C}$

$$S_{BC} = 8.1 + 6 - 13.5 = 0.6$$

Clarke and Wright Savings Method

Clarke and Wright Savings Ranking & Tour Construction

Rank	Saving	From	To	Tour	Remarks
1	9.5	D	E	[1] **A-D-E-A**	Initiate new tour
2	8.3	C	D	[1] **A-C-D-E-A**	Insert C between A and D
3	7.6	C	E		
4	6.7	H	I	[1] **A-C-D-E-A** [2] **A-H-I-A**	Initiate new tour
5	5.8	D	I		
6	5.8	F	G	[1] **A-C-D-E-A** [2] **A-H-I-A** [3] **A-F-G-A**	Initiate new tour
7	5.6	D	H		
8	5.2	B	G	[1] **A-C-D-E-A** [2] **A-H-I-A** [3] **A-F-G-B-A**	Insert B between G and A
9	5.0	E	H	[1] **A-C-D-E-H-I-A** [2] **A-F-G-B-A**	Join E and H and merge the subtours
10	4.7	E	I		
11	4.4	C	I		
12	3.8	C	H		
13	3.8	D	F		
14	3.3	B	F		
15	3.0	E	F		
16	2.8	B	I	[1] **A-C-D-E-H-I-B-G-F-A**	Join B and I and merge the subtours
17	2.7	D	G		
18	2.6	B	H		

What are the 3 conditions for nodes linking?

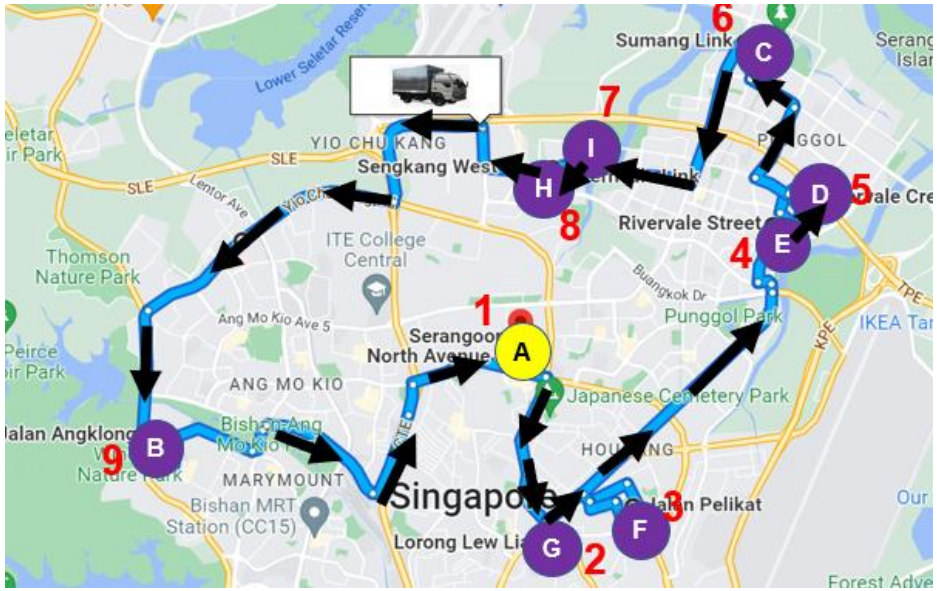
Route:

A -> C -> D -> E -> H -> I -> B -> G -> F -> A

From	To	Distance
A	C	6
C	D	3.7
D	E	1.7
E	H	4.2
H	I	1.3
I	B	9.3
B	G	6.7
G	F	2.3
F	A	4.3

Total Dist. 39.5 km

Nearest Neighbor Vs C&W



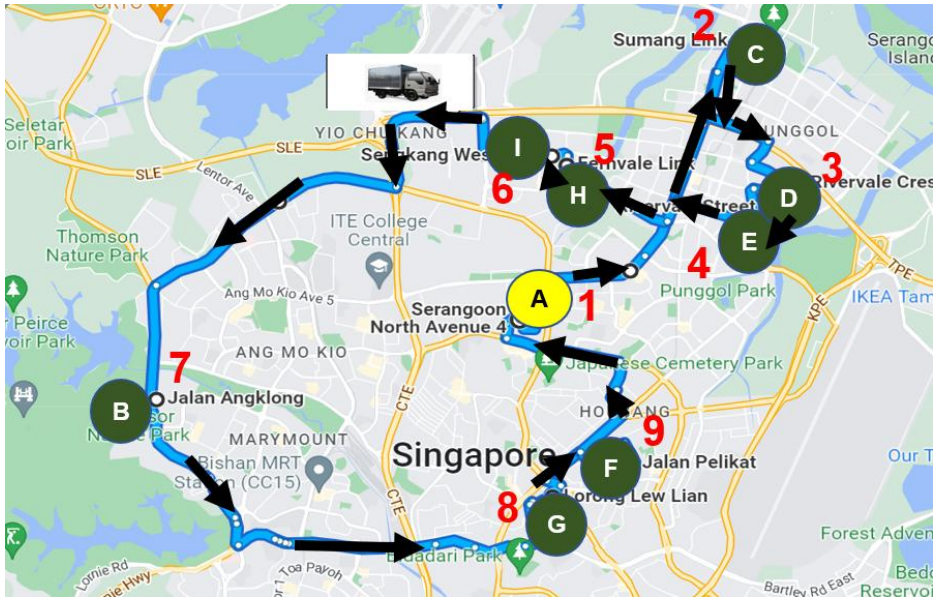
From	To	Distance
A	G	3.8
G	F	2.3
F	E	6.5
E	D	1.7
D	C	3.7
C	I	5.6
I	H	1.3
H	B	9.5
B	A	8.1

Total Dist. 42.5 km

Nearest Neigbourhood

Route:

A -> G -> F -> E -> D -> C -> I -> H -> B -> A



From	To	Distance
A	C	6
C	D	3.7
D	E	1.7
E	H	4.2
H	I	1.3
I	B	9.3
B	G	6.7
G	F	2.3
F	A	4.3

Total Dist. 39.5 km

Clarke & Wright Savings

Route:

A -> C -> D -> E -> H -> I -> B -> G -> F -> A

C&W Savings method provides shorter distance

With Vehicle Capacity Constraint

Clarke and Wright Savings Method

Rank	Saving	From	To	Tour	Remarks	Vehicle load	Comments
1	9.5	D	E	[1] **A-D-E-A**	Initiate new tour	[1] 150 + 200 = 350	
2	8.3	C	D	[1] **A-C-D-E-A**	Insert C between A and D	[1] 350 + 100 = 450	
3	7.6	C	E				
4	6.7	H	I	[1] **A-C-D-E-A** [2] **A-H-I-A**	Initiate new tour	[1] 450 [2] 180 + 90 = 270	
5	5.8	D	I				
6	5.8	F	G	[1] **A-C-D-E-A** [2] **A-H-I-A** [3] **A-F-G-A**	Initiate new tour	[1] 450 [2] 270 [3] 80 + 100 = 180	
7	5.6	D	H				
8	5.2	B	G	[1] **A-C-D-E-A** [2] **A-H-I-A** [3] **A-F-G-B-A**	Insert B between G and A	[1] 450 [2] 270 [3] 180 + 150 = 330	
9	5.0	E	H	[1] **A-C-D-E-A** [2] **A-H-I-A** [3] **A-F-G-B-A**		[1] 450 [2] 270 [3] 330	If E and H are joined, A-C-D-E-H-I-A load will be 720, which exceeds veh capacity
10	4.7	E	I				
11	4.4	C	I				
12	3.8	C	H				
13	3.8	D	F				
14	3.3	B	F				
15	3.0	E	F	[1] **A-C-D-E-A** [2] **A-H-I-A** [3] **A-F-G-B-A**		[1] 450 [2] 270 [3] 330	If E and F are joined, A-C-D-E-F-G-B-A load will be 800, which exceeds veh capacity
16	2.8	B	I	[1] **A-C-D-E-A** [2] **A-H-I-B-G-F-A**		[1] 450 [2] 270 + 330 = 600	Both tours' loads are within van capacity
17	2.7	D	G				

Customer	Delivery Load (kg)
B	150
C	100
D	150
E	200
F	80
G	100
H	180
I	90
Sum	1050

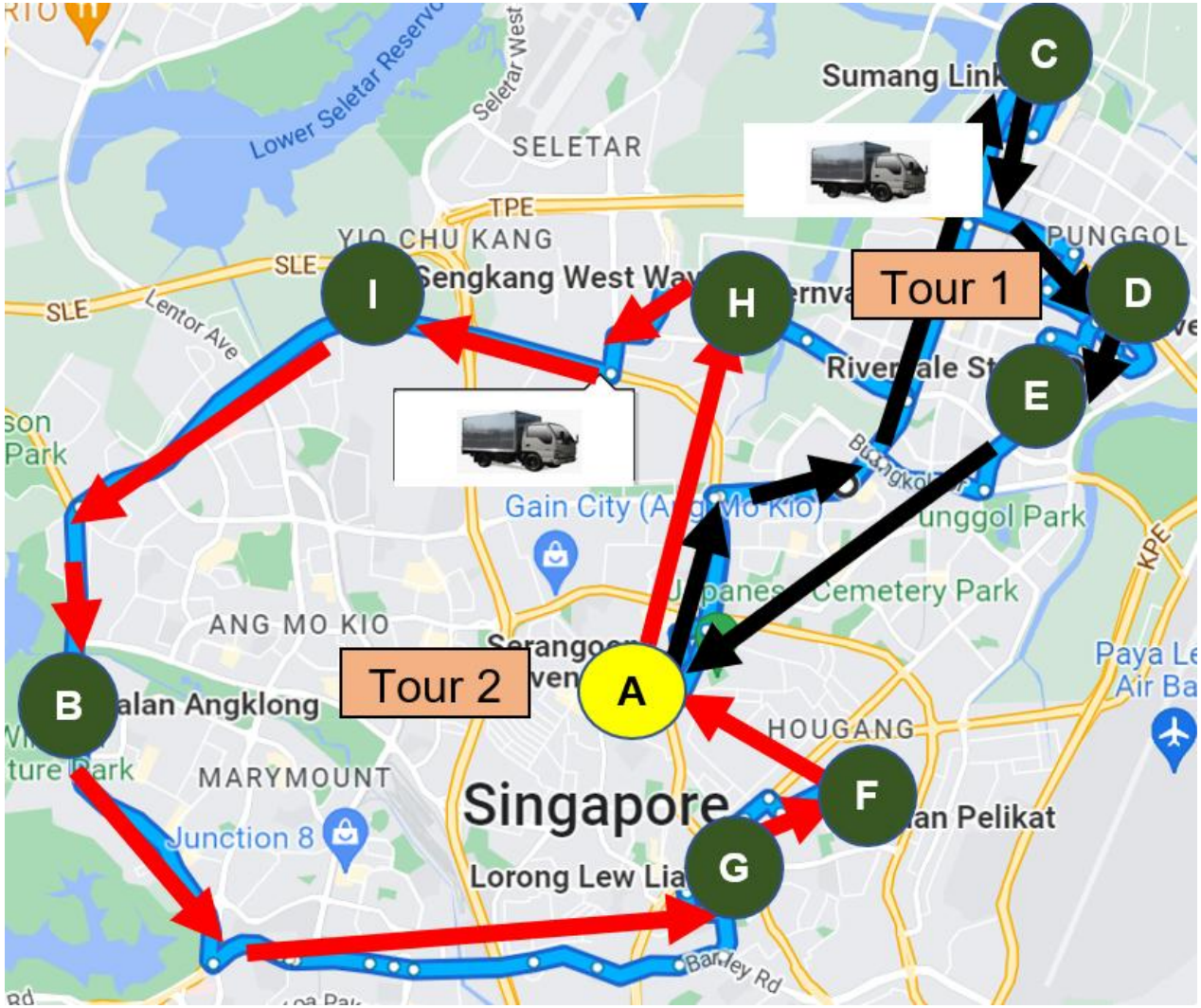
Tour [1]

From	To	Distance (km)	Delivery Load (kg)
A	C	6	100
C	D	3.7	150
D	E	1.7	200
E	A	5.2	
	Total	**16.6 km**	**450 kg**

Tour [2]

From	To	Distance (km)	Delivery Load (kg)
A	H	4	180
H	I	1.3	90
I	B	9.3	150
B	G	6.7	100
G	F	2.3	80
F	A	4.3	
	Total Dist.	**27.9 km**	**600 kg**

Route Visualization



Tour	Load (kg)	Distance (km)
[1] A-C-D-E-A	450	16.6
[2] A-H-I-B-G-F-A	600	27.9
Total	1050	44.5

Summary

- Clarke & Wright Savings Method will normally provide a better routing solution as compared with the Nearest Neighbour method. However, both methods do not guarantee optimal solutions.
- Clarke & Wright Savings Method is more tedious as compared to Nearest Neighbour method, which makes it more difficult to implement in a real application.
- If delivery time window is imposed at a certain location, the location's sequence would have to be scheduled to meet the time window constraint. For example, if delivery must be made at node F first, during tour construction, avoid inserting a node between Depot and node F.
- Delivery failure or unexpected delay in delivery can arise from a variety of reasons, such as customers not present during delivery, wrong address or wrong contact number so driver needs to do clarification, intercom is not working (e.g, for condos), language barrier in reading instructions for some drivers, traffic jam, bad weather, waiting for parking, rejection by customers, etc. There should be contingency plans for failed deliveries.

Lesson 20 Solution

Minimum cost flow problem

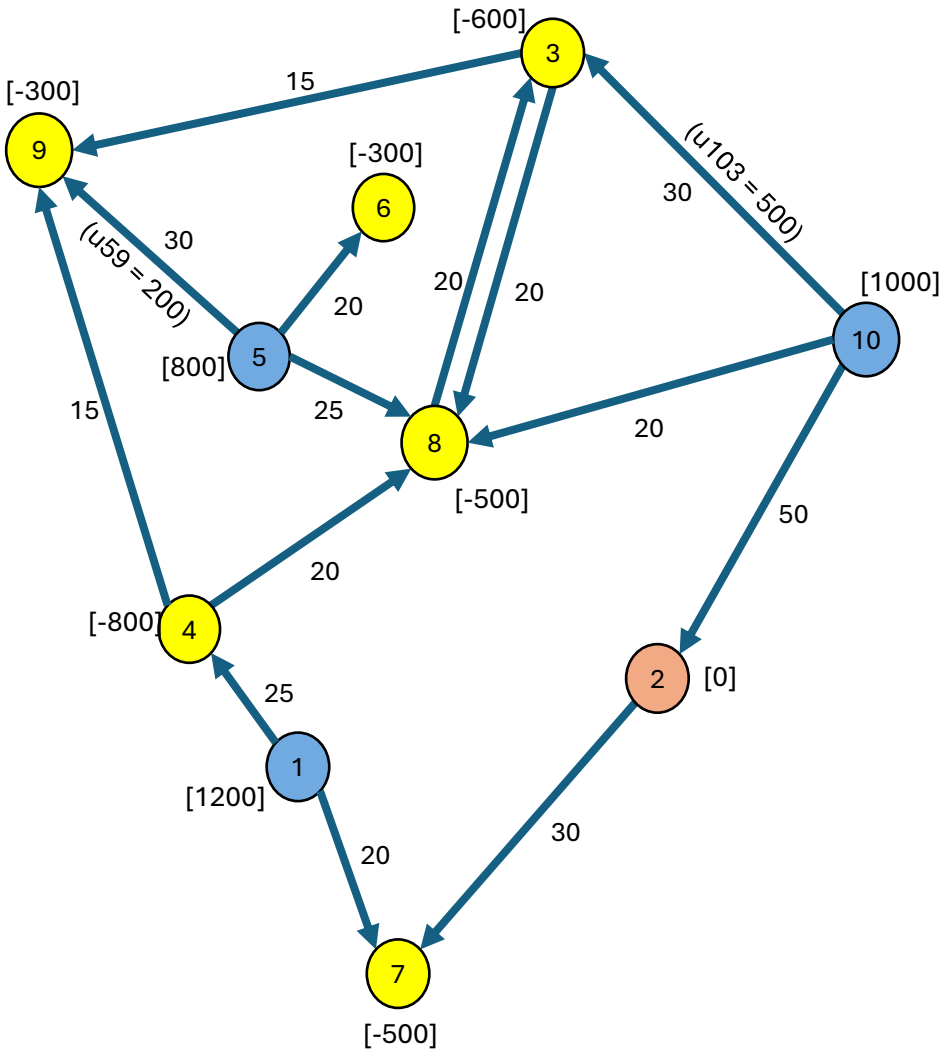
Learning Outcomes

- Explain that a distribution network problem deals with the distribution of goods from ‘sources’ to ‘destinations’ in a network with transhipment points and may consider cost of shipment and flow capacity.

- Formulate LP models for the minimum cost flow problem and shortest path problem.

- Solve LP models using Excel Solver for the minimum cost flow problem and shortest path problem.

Formulate a Minimum Cost Flow LP Model



Node	Location	Type of Node	Net flow
1	Singapore	Supply	1200
2	Bandar Seri Begawan	Transhipment	0
3	Hanoi	Demand	-600
4	Kuala Lumpur	Demand	-800
5	Bangkok	Supply	800
6	Vientiane	Demand	-300
7	Jakarta	Demand	-500
8	Phnom Penh	Demand	-500
9	Naypyidaw	Demand	-300
10	Manila	Supply	1000

Decision Variables & Objective Function

Decision Variables:

Let X_{ij} be the **number of units of toys t**o be shipped from node *i* to node *j*. In total, there are 14 **decision variables (same as the number of arcs)**:

X_{14}, $X_{17,}$ X_{27}, X_{38}, X_{39}, X_{48}, $X_{49,}$ X_{56}, X_{58}, X_{59}, X_{83}, X_{102}, X_{103}, X_{108}

Objective function:

Minimize total shipment cost, Z =

$25X_{14} + 20X_{17} + 30X_{27} + 20X_{38} + 15X_{39} + 20X_{48} + 15X_{49} + 20X_{56} + 25X_{58} + 30X_{59} + 20X_{83} + 50X_{102} + 30X_{103} + 20X_{108}$

Net Flow Constraints

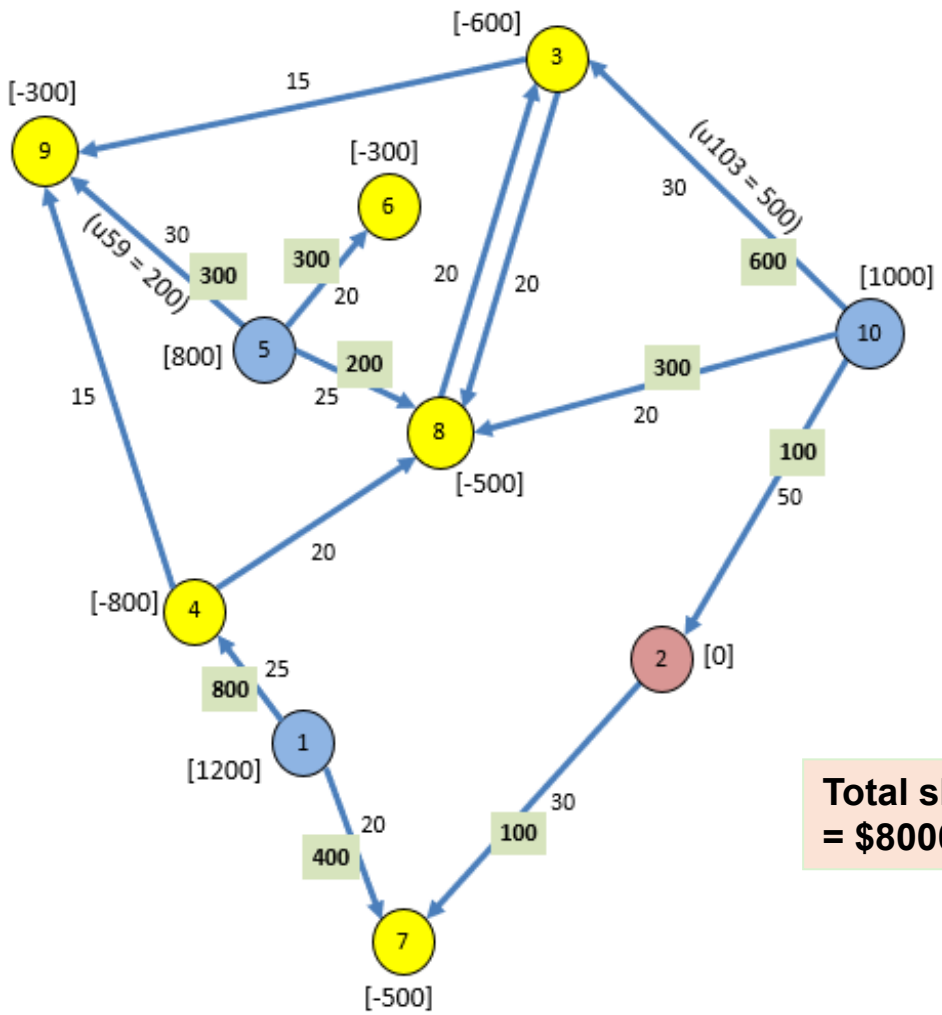
Subject to: Net Flow Constraints at Nodes

Supply Node 1:	x14 + x17	=	1000
Supply Node 5:	x56 + x58 + x59	=	900
Supply Node 10:	x102 + x103 + x108	=	1200
Transhipment Node 2:	x27 - x102	=	0
Demand Node 3:	x38 + x39 - x83 - x103	=	-700
Demand Node 4:	x48 + x49 - x14	=	-800
Demand Node 6:	- x56	=	-200
Demand Node 7:	- x17 - x27	=	-500
Demand Node 8:	x83 - x38 - x48 - x58 - x108	=	-500
Demand Node 9:	- x39 - x49 - x59	=	-400

Non-negativity constraint:

All X_{ij} >= 0

Solver Solution (No Arc Capacity Constraint)



Total shipment cost = $80000

Arc	Units
x14	800
x17	400
x27	100
x38	0
x39	0
x48	0
x49	0
x56	300
x58	200
x59	300
x83	0
x102	100
x103	600
x108	300

Arc Capacity Constraints

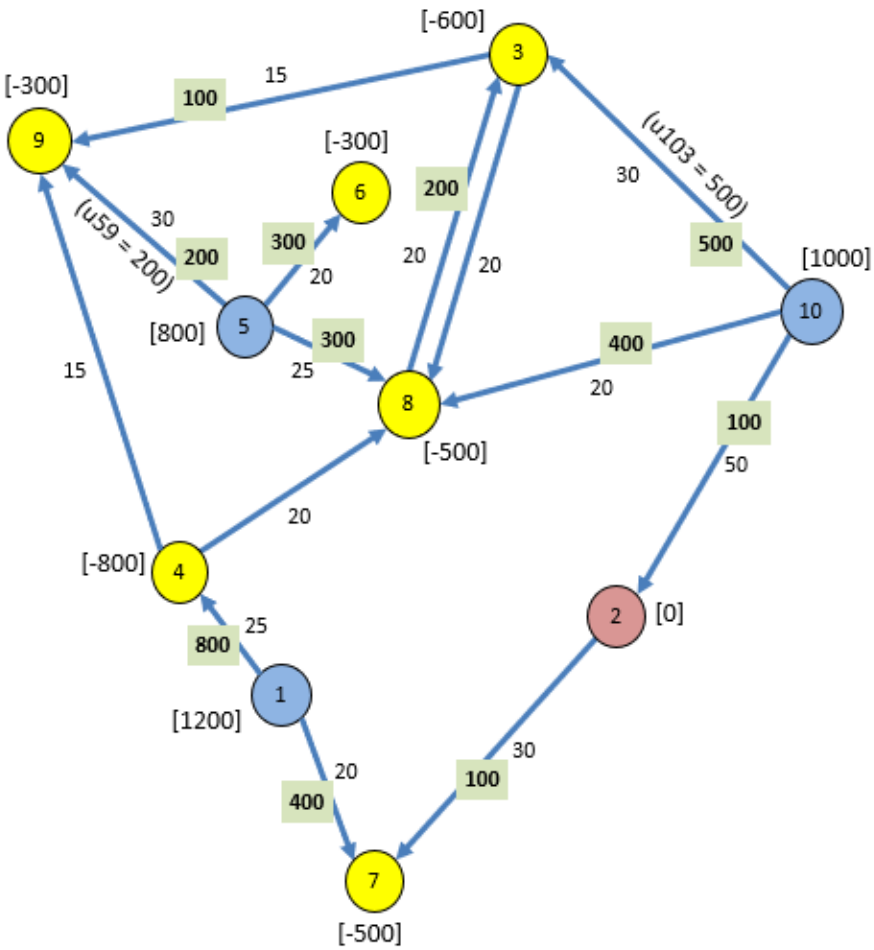
Shipping capacity is restricted to a maximum of 200 & 500 toy units from Bangkok (5) to Naypyidaw (9) and Manila (10) to Hanoi (3) respectively.

Arc capacity or Upper-Bound constraints

$X_{59} <= 200$

$X_{103} <= 500$

Solver Solution (with Arc Capacity Constraints)



Arc	Units
x14	800
x17	400
x27	100
x38	0
x39	100
x48	0
x49	0
x56	300
x58	300
x59	200
x83	200
x102	100
x103	500
x108	400

Total shipment cost = $84000

Total shipment cost is higher compared with no capacity constraint. The shipment capacity constraints result in the distribution becoming less optimal.

Total Supply ≠ Total Demand

How would the constraints in the LP model change if:

a) The plant at Singapore can only supply 1000 units of toys instead of 1200
b) The demand at Phnom Penh is reduced to 400 from 500

a) **Total supply < total demand**
→ All supplies are sent out whereas some demands are not fulfilled.

Change all demand node constraint signs to '>=' (because RHS values are negative).

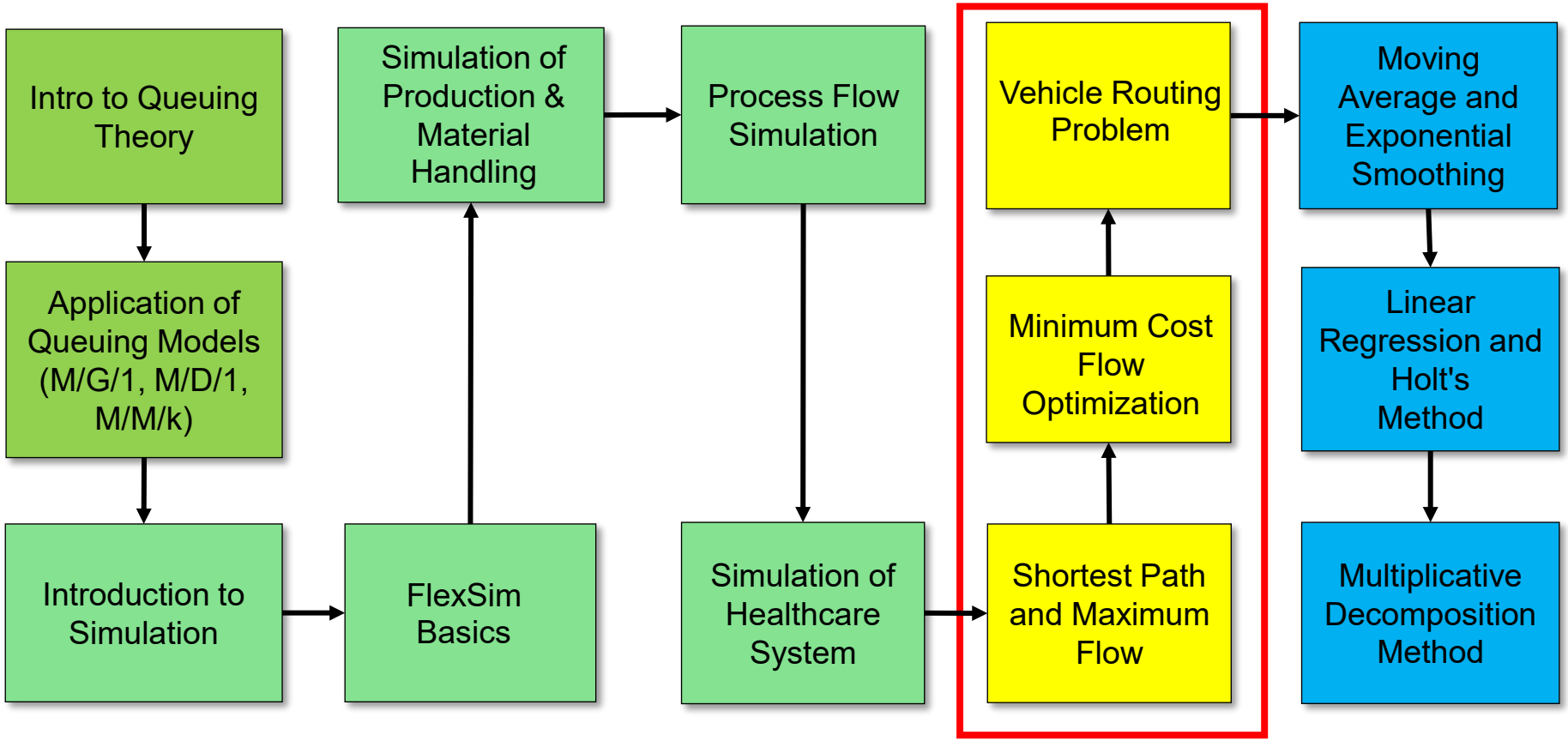
Demand Node 3:	x38 + x39 - x83 - x103	>=	-600
Demand Node 4:	x48 + x49 - x14	>=	-800
Demand Node 6:	- x56	>=	-300
Demand Node 7:	- x17 - x27	>=	-500
Demand Node 8:	x83 - x38 - x48 - x58 - x108	>=	-500
Demand Node 9:	- x39 - x49 - x59	>=	-300
Supply Node 1:	x14 + x17	=	1000

b) **Total supply > total demand**
→ Some supplies are not sent out whereas all demands are met.

Change all supply node constraint signs should be to '<='

Supply Node 1:	x14 + x17	<=	1200
Supply Node 5:	x56 + x58 + x59	<=	800
Supply Node 10:	x102 + x103 + x108	<=	1000
Demand Node 8:	x83 - x38 - x48 - x58 - x108	=	-400

Summary

- The minimum cost flow problem holds a central position in the network optimization models.
- The transportation problem, shortest path problem, and maximum flow problem are all special cases of the minimum cost flow problem.
- One special application of the minimum cost flow problem is the distribution network problem.
- The distribution network problem is to find the minimum cost shipment arrangements distributing products from ‘sources’ to ‘destinations’ considering both transshipment points and flow capacity in a transportation network.
- Flow networks can also be used to model liquids flowing through pipes, parts through assembly lines, current through electrical networks, information through communication networks.

E211 Progression